Experimental Table:

Data Table 01: Verification of KVL

OBS	V(V)	V_1(V)		V_2(V)		V_3(V)		$V = V_1 + V_2 + V_3$ (V)	
		Cal	Mea	Cal	Mea	Cal	Mea	Cal	Mea
1.	4	1.09	1.0	2.40	2.4	0.51	0.4	4.00	3.8
2.	6	1.63	1.6	3.60	3.6	0.77	0.7	6.00	5.9
3.	8	2.18	2.1	4.80	4.8	1.02	1.0	8.00	7.9

Data Table 02: Verification of KCL

OBS	V(V)	I_1(mA)		I_2(mA)		$I = I_1 + I_2$ (mA)	
		Cal	Mea	Cal	Mea	Cal	Mea
1.	4	4.00	4.0	1.82	1.8	5.82	5.8
2.	6	6.00	6.1	2.73	2.7	8.73	8.8
3.	8	8.00	8.1	3.64	3.6	11.64	11.7

Calculations:

Given: R_1 = 1000 Ω, R_2 = 2200 Ω, R_3 = 470 Ω

KVL

OBS 1:

$$V1 = \frac{V \times R1}{R1 + R2 + R3} = \frac{4 \times 1000}{1000 + 2200 + 470} = 1.09\ V$$

$$V2 = \frac{V \times R2}{R1 + R2 + R3} = \frac{4 \times 2200}{1000 + 2200 + 470} = 2.40\ V$$

$$V3 = \frac{V \times R3}{R1 + R2 + R3} = \frac{4 \times 470}{1000 + 2200 + 470} = 0.51\ V$$

$V_{mea} = V_1 + V_2 + V_3 = 1.0 + 2.4 + 0.4 = 3.8\ V$
$V_{cal} = V_1 + V_2 + V_3 = 1.09 + 2.40 + 0.51 = 4.00\ V$

OBS 2:

$$V1 = \frac{V \times R1}{R1 + R2 + R3} = \frac{6 \times 1000}{1000 + 2200 + 470} = 1.63\ V$$

$$V2 = \frac{V \times R2}{R1 + R2 + R3} = \frac{6 \times 2200}{1000 + 2200 + 470} = 3.60\ V$$

$$V3 = \frac{V \times R3}{R1 + R2 + R3} = \frac{6 \times 470}{1000 + 2200 + 470} = 0.77\ V$$

$V_{mea} = V_1 + V_2 + V_3 = 1.6 + 3.6 + 0.7 = 5.9\ V$
$V_{cal} = V_1 + V_2 + V_3 = 1.63 + 3.60 + 0.77 = 6.00\ V$

OBS 3:

$$V1 = \frac{V \times R1}{R1 + R2 + R3} = \frac{8 \times 1000}{1000 + 2200 + 470} = 2.18\ V$$

$$V2 = \frac{V \times R2}{R1 + R2 + R3} = \frac{8 \times 2200}{1000 + 2200 + 470} = 4.80\ V$$

$$V3 = \frac{V \times R3}{R1 + R2 + R3} = \frac{8 \times 470}{1000 + 2200 + 470} = 1.02\ V$$

$V_{mea} = V_1 + V_2 + V_3 = 2.1 + 4.8 + 1.0 = 7.9\ V$
$V_{cal} = V_1 + V_2 + V_3 = 2.18 + 4.80 + 1.02 = 8.00\ V$

KCL

OBS 1:

$$I_1 = \frac{I \times R2}{R1 + R2} = \frac{5.82 \times 2200}{1000 + 2200} = 4.00\ mA$$

$$I_2 = \frac{I \times R1}{R1 + R2} = \frac{5.82 \times 1000}{1000 + 2200} = 1.82\ mA$$

$I_{mea} = I_1 + I_2 = 4.0 + 1.8 = 5.8\ mA$
$I_{cal} = I_1 + I_2 = 4.00 + 1.82 = 5.82\ mA$

OBS 2:

$$I_1 = \frac{I \times R2}{R1 + R2} = \frac{8.73 \times 2200}{1000 + 2200} = 6.00\ mA$$

$$I_2 = \frac{I \times R1}{R1 + R2} = \frac{8.73 \times 1000}{1000 + 2200} = 2.73\ mA$$

$I_{mea} = I_1 + I_2 = 6.1 + 2.7 = 8.8\ mA$
$I_{cal} = I_1 + I_2 = 6.00 + 2.73 = 8.73\ mA$

OBS 3:

$$I_1 = \frac{I \times R2}{R1 + R2} = \frac{11.64 \times 2200}{1000 + 2200} = 8.00 \text{ mA}$$

$$I_2 = \frac{I \times R1}{R1 + R2} = \frac{11.64 \times 1000}{1000 + 2200} = 3.64 \text{ mA}$$

$I_{mea} = I_1 + I_2 = 8.1 + 3.6 = 11.7$ mA

$I_{cal} = I_1 + I_2 = 8.00 + 3.64 = 11.64$ mA